

A Five-Cell Counterexample to the E-Bolt Error Formula

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Status. Candidate counterexample, likely valid, to Theorem 3.8 of Ismailov's notes [1] and to the subsequent e-bolt conjecture under the source's explicit identification of the e-bolts of that polygon. The counterexample uses the literal printed definition of $M(Q)$: the mixed increment condition is required for each rectangle contained as a set in Q .

1 Source statement

The notes define, for a closed bolt $p = (p_1, \dots, p_{2n})$,

$$l(f, p) = \frac{1}{2n} \sum_{k=1}^{2n} (-1)^{k-1} f(p_k).$$

For a polygon Q , $M(Q)$ consists of the continuous functions satisfying

$$f(x_1, y_1) + f(x_2, y_2) - f(x_1, y_2) - f(x_2, y_1) \geq 0 \tag{1}$$

for every axis-parallel rectangle $[x_1, x_2] \times [y_1, y_2] \subset Q$.

Theorem 3.8, on article page 140 of the notes, takes

$$\begin{aligned} R_1 &= [a_1, a_4] \times [b_1, b_2], \\ R_2 &= [a_1, a_2] \times [b_2, b_3], \\ R_3 &= [a_3, a_4] \times [b_2, b_3], \quad Q = R_1 \cup R_2 \cup R_3, \end{aligned}$$

and asserts that every $f \in M(Q)$ satisfies

$$E(f, Q) = \max\{|l(f, r)|, |l(f, r_{12})|, |l(f, r_{13})|\}, \tag{2}$$

where the three bolts are formed by the vertices of the outer rectangle, $R_1 \cup R_2$, and $R_1 \cup R_3$, respectively.

$$Q = \bigcup_{i=1}^3 R_i, \quad \text{where}$$

$R_1 = [a_1, a_2] \times [b_1, b_2]$, $R_2 = [a_2, a_3] \times [b_1, b_2]$, $R_3 = [a_3, a_4] \times [b_1, b_2]$, $R_4 = [a_2, a_3] \times [b_2, b_3]$. Let $f \in M(Q)$. Then the following formula holds

$$E(f, Q) = \max \{ |l(f, q)|, |l(f, r_{123})|, |l(f, r_{124})|, |l(f, r_{234})|, |l(f, r_{24})| \},$$

where $q, r_{123}, r_{124}, r_{234}, r_{24}$ are closed bolts formed by the vertices of the polygons $Q, R_1 \cup R_2 \cup R_3, R_1 \cup R_2 \cup R_4, R_2 \cup R_3 \cup R_4$ and $R_2 \cup R_4$, respectively.

Theorem 3.8. Let $a_1 < a_2 < a_3 < a_4$, $b_1 < b_2 < b_3$ and Q be an octagon of the following form

$$Q = \bigcup_{i=1}^3 R_i,$$

where $R_1 = [a_1, a_4] \times [b_1, b_2]$, $R_2 = [a_1, a_2] \times [b_2, b_3]$, $R_3 = [a_3, a_4] \times [b_2, b_3]$. Let $f \in M(Q)$. Then

$$E(f, Q) = \max \{ |l(f, r)|, |l(f, r_{12})|, |l(f, r_{13})| \},$$

where r, r_{12}, r_{13} are closed bolts formed by the vertices of the polygons $R = [a_1, a_4] \times [b_1, b_3]$, $R_1 \cup R_2, R_1 \cup R_3$, respectively.

Although the closed bolts maximization process can be applied to bolts of an arbitrary polygon, some combinatorial difficulties arise when grouping values at points of maximized bolts (bolts obtained after the maximization process, see (3.15)-(3.18)). While we do not know a complete answer to this problem, we can describe points of a polygon F with which points of maximized bolts coincide and state a conjecture concerning the approximation error.

Let $F = A_1 A_2 \dots A_{2n}$ be any polygon with sides parallel to the coordinate axes. The vertices $A_1, A_2, \dots, A_{2n}$ in the given order form a closed bolt, which we denote by r_F . By $[r_F]$ denote the length of r_F . In our case, $[r_F] = 2n$.

Definition 3.2. Let F and S be polygons with sides parallel to the coordinate axes. We say that the closed bolt r_F is an *e-bolt* (extended bolt) of S

Definition 3.2 introduces e-bolts. The paragraph immediately following it states explicitly that the polygon in Theorem 3.8 has exactly the three e-bolts r, r_{12}, r_{13} . The notes then conjecture, for every axis-parallel polygon S and $f \in M(S)$,

$$E(f, S) = \max_{h \in S^E} |l(f, h)|. \quad (3)$$

if $r_F \subset S$ and there does not exist any polygon F' such that $F \subset F'$, $r_{F'} \subset S$, $[r_{F'}] \leq [r_F]$.

For example, in Theorem 3.8 the octagon Q has 3 e -bolts. They are r, r_{12} and r_{13} . In Theorem 3.7, the octagon Q has 5 e -bolts, which are $q, r_{123}, r_{124}, r_{234}$ and r_{24} . The polygon $S_{2n} = \bigcup_{i=1}^{n-1} R_i$, where $R_i = [a_i, a_{i+1}] \times [b_1, b_{n+1-i}]$, $i = \overline{1, n-1}$, $a_1 < a_2 < \dots < a_n, b_1 < b_2 < \dots < b_n$ has exactly $2^{n-1} - 1$ e -bolts. It is not difficult to observe that the set of points of a closed bolt obtained after the maximization process is a subset of the set of points of all e -bolts. This condition and Theorems 2.5-2.8 justify the statement of the following conjecture:

Let S be any polygon with sides parallel to the coordinate axes and $f \in M(S)$. Then

$$E(f, S) = \max_{h \in S^E} \{|l(f, h)|\},$$

where S^E is a set of all e -bolts of the polygon S .

2 Proof Intuition

The missing top-middle cell of the five-cell U-shaped polygon is invisible to the definition of $M(Q)$: no rectangle contained in Q enters that cell. We therefore place mixed density $+1$ in an occupied lower-left cell and mixed density -1 in the missing cell. The restriction to Q remains in $M(Q)$, because every admissible rectangle sees only the nonnegative density. The negative density nevertheless cancels values on the long e -bolts used in (2). A short four-point bolt around the positive cell has value $1/4$, while each of the three displayed e -bolts has absolute value at most $1/6$. Since every closed-bolt functional annihilates $\varphi(x) + \psi(y)$, the short bolt forces $E(f, Q) \geq 1/4$ and contradicts the claimed formula.

3 Counterexample

Theorem 1. *Theorem 3.8 of [1], with the printed definition of $M(Q)$, is false. More precisely, there is an explicit continuous $f \in M(Q)$ for which the right-hand side of (2) equals $1/6$, whereas*

$$E(f, Q) \geq \frac{1}{4}.$$

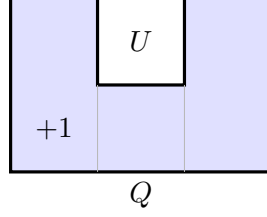
Proof. Choose

$$a_1 = 0, \quad a_2 = 1, \quad a_3 = 2, \quad a_4 = 3, \quad b_1 = 0, \quad b_2 = 1, \quad b_3 = 2.$$

Thus

$$Q = ([0, 3] \times [0, 1]) \cup ([0, 1] \times [1, 2]) \cup ([2, 3] \times [1, 2]). \quad (4)$$

This is the five-cell U-shaped polygon shown below; its missing cell is $U = (1, 2) \times (1, 2)$.



On the bounding rectangle $[0, 3] \times [0, 2]$, define

$$f(x, y) = \min\{x, 1\} \min\{y, 1\} - \min\{(x-1)_+, 1\} (y-1)_+, \quad (5)$$

and restrict f to Q . This function is continuous and piecewise bilinear. Its mixed derivative, in the almost-everywhere and distributional sense, is

$$\partial_x \partial_y f = \mathbf{1}_{(0,1) \times (0,1)} - \mathbf{1}_{(1,2) \times (1,2)}. \quad (6)$$

Let $T = [x_1, x_2] \times [y_1, y_2] \subset Q$ be any axis-parallel rectangle. Its interior is disjoint from the missing cell U . Applying the two-variable fundamental theorem of calculus to the Lipschitz, piecewise-bilinear function (5) gives

$$\begin{aligned} & f(x_1, y_1) + f(x_2, y_2) - f(x_1, y_2) - f(x_2, y_1) \\ &= \iint_T \partial_x \partial_y f = |T \cap ((0, 1) \times (0, 1))| \geq 0. \end{aligned} \quad (7)$$

Hence $f \in M(Q)$.

We now evaluate the three bolts in Theorem 3.8. Orienting them as

$$\begin{aligned} r &= ((0, 0), (3, 0), (3, 2), (0, 2)), \\ r_{12} &= ((0, 0), (3, 0), (3, 1), (1, 1), (1, 2), (0, 2)), \\ r_{13} &= ((0, 0), (3, 0), (3, 2), (2, 2), (2, 1), (0, 1)), \end{aligned}$$

formula (5) gives the corresponding value lists

$$(0, 0, 0, 0), \quad (0, 0, 1, 1, 1, 0), \quad (0, 0, 0, 0, 1, 0).$$

Therefore

$$l(f, r) = 0, \quad l(f, r_{12}) = \frac{1}{6}, \quad l(f, r_{13}) = \frac{1}{6}, \quad (8)$$

so the right-hand side of (2) is $1/6$.

Finally consider the closed bolt

$$q = ((0, 0), (1, 0), (1, 1), (0, 1)) \subset Q.$$

Its values are $(0, 0, 1, 0)$, whence

$$l(f, q) = \frac{1}{4}. \quad (9)$$

For every pair of continuous univariate functions φ, ψ , the alternating sum on a closed bolt cancels coordinate by coordinate, so $l(\varphi + \psi, q) = 0$. Consequently,

$$\frac{1}{4} = |l(f, q)| = |l(f - \varphi - \psi, q)| \leq \|f - \varphi - \psi\|_{C(Q)}.$$

Taking the infimum over $\varphi + \psi$ yields $E(f, Q) \geq 1/4$. This contradicts (2). $\square$

Corollary 1. *Under the source’s explicit statement that the polygon (4) has exactly the three e-bolts r, r_{12}, r_{13} , the e-bolt conjecture (3) is false.*

Proof. For this Q and f , equations (8)–(9) give

$$\max_{h \in Q^E} |l(f, h)| = \frac{1}{6} < \frac{1}{4} \leq E(f, Q).$$

$\square$

4 Verification, conventions, and scope

The proof uses no numerical approximation. The accompanying checker performs all displayed calculations over exact rational numbers and also tests (1) on every rational-grid rectangle with mesh $1/8$ that is contained in Q . The command

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python code/check_five_cell_counterexample.py
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returns the three e-bolt values $0, 1/6, 1/6$, the short-bolt value $1/4$, and no negative mixed increment among the tested rectangles. The finite grid test is only a sanity check; the proof for all rectangles is (6)–(7).

There is an important convention issue for human review. The source’s printed condition is set containment of the whole rectangle in Q , and that is the condition proved in (7). If an unstated stronger convention were intended in which (1) is required whenever all four rectangle vertices lie in Q , then the missing cell itself would be tested and (5) would not qualify. That stronger convention is not what the displayed definition says.

Theorem 1 does not depend on resolving any ambiguity in the general definition of e-bolts: it directly contradicts the three-term formula printed in Theorem 3.8. Corollary 1 additionally uses the notes’ own sentence that these three bolts are exactly the e-bolts of this polygon.

5 Novelty check

The run indexes were searched for arXiv:2005.14125, the paper title, Theorem 3.8, e-bolts, extended bolts, and the 2006 article title; no prior packet or attempt addressing this formula was found. Bounded web searches on 22 July 2026 used the exact phrases *e-bolts*, *extended bolt*, *On error formulas for approximation by sums of univariate functions*, and *Conjecture 3.4*, together with Ismailov’s name and terms such as counterexample and polygon. They found the 2006 source article [2], the 2020 notes [1], and bibliographic mirrors, but no later proof, counterexample, or discussion of this obstruction. Novelty confidence is moderate pending specialist and author review, especially because the result exposes a possible mismatch between the printed rectangle condition and an intended stronger one.

6 Human-review recommendation

Review first as a literal counterexample to Theorem 3.8. The decisive checks are: (i) the source definition of $M(Q)$ requires the whole rectangle to be contained in Q ; (ii) the three source bolts are ordered as in (8); and (iii) every closed-bolt functional gives the standard lower bound for $E(f, Q)$. If these pass, the theorem is false as printed. Then verify the source's explicit assertion that $Q^E = \{r, r_{12}, r_{13}\}$ to accept the corollary for the conjecture.

References

- [1] V. E. Ismailov, *Notes on ridge functions and neural networks*, arXiv:2005.14125, 2020; Theorem 3.8 and Definition 3.2 on article pages 140–141.
- [2] V. E. Ismailov, *On error formulas for approximation by sums of univariate functions*, International Journal of Mathematics and Mathematical Sciences **2006** (2006), Article ID 65620, 11 pp., doi:10.1155/IJMMS/2006/65620.